Chapter 1 Gaussian Process

Introduction

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1.1 Stochastic Process

Definition 1.1 (Stochastic Process)

A stochastic process *is a collection of random variables (possibly infinitely many) indexed by a set* $\mathcal{I}$.

The choice of index set $\mathcal{I}$ is flexible; common examples include the natural numbers $\mathbb{N}$ or the real line $\mathbb{R}$. We may freely draw sample pairs (x_i, y_i) with $y_i \sim p(x_i)$, and in the limit of infinitely many samples the joint distribution of these pairs becomes determined. Conversely, we may also characterize a stochastic process directly by specifying the joint distributions of the random variables $\{y_i\}_{i\in\mathcal{I}}$ for all finite index subsets.

Definition 1.2 (Markov Random Process)

A Markov random process $\{y_i\}_{i\in\mathbb{N}}$ *is a stochastic process satisfying*

$$p(y_{t+1} \mid y_1, \ldots, y_t) = p(y_{t+1} \mid y_t), \qquad \forall t \in \mathbb{N}. \tag{1.1}$$

Definition 1.3 (Bernoulli Random Process)

A Bernoulli random process $\{y_i\}_{i\in\mathbb{N}}$ *is a stochastic process satisfying*

$$p(y_1, \ldots, y_n) = \prod_{i=1}^{n} p(y_i), \tag{1.2}$$

meaning that the random variables are mutually independent, and each $y_i \sim \mathcal{B}(1, p_i)$ *follows a Bernoulli distribution.*

Definition 1.4 (Multivariate Gaussian)

For $x \in \mathbb{R}^d$*, we say that* x *follows a* multivariate Gaussian distribution *if*

$$p(x) = \frac{1}{(2\pi)^{d/2}} \frac{1}{\det(\Sigma)^{1/2}} \exp\left(-\frac{1}{2}(x-\mu)^\top \Sigma^{-1}(x-\mu)\right), \tag{1.3}$$

where $\mu \in \mathbb{R}^d$ *is the mean vector and* $\Sigma \in \mathbb{R}^{d\times d}$ *is a positive-definite symmetric covariance matrix. We denote this distribution by* $x \sim \mathcal{N}(\mu, \Sigma)$.

It is easy to verify that the multivariate Gaussian reduces to the standard one-dimensional Gaussian distribution when $d = 1$. Moreover, when the covariance matrix Σ is diagonal, the multivariate Gaussian factorizes into the product of d independent one-dimensional Gaussian distributions.

Remark

More detailed mathematical treatment can be found in Pattern Recognition and Machine Learning *(Bishop, Sec. 2.3).*

Theorem 1.1

Any affine (or linear) transformation of a Gaussian random variable remains Gaussian. ♡

The above property can in fact be taken as an equivalent way to *define* the Gaussian distribution, due to the equivalence between the standard density-based definition and this transformation property.

Note

This property also extends to the case where the covariance matrix Σ is degenerate.

Definition 1.5 (Gaussian Process)

A Gaussian Process *(GP) is a collection of random variables $\{f(x)\}_{x\in\mathcal{X}}$ indexed by an input space $\mathcal{X}$, such that every finite subcollection*

$$\big(f(x_1), f(x_2), \ldots, f(x_n)\big) \tag{1.4}$$

is jointly multivariate Gaussian for all choices of $x_1, \ldots, x_n \in \mathcal{X}$. ♣

A natural question arises: to specify a Gaussian process, do we need to provide a different set of parameters for every collection of n random variables? If that were the case, the notion of a Gaussian process would lose its meaning, since incorporating more sample points would require specifying exponentially many parameters, which is an infeasible task when n is large.

Fortunately, Gaussian processes enjoy the desirable closure property under linear transformations. This guarantees that adding new random variables does not affect the parameters of existing ones, because the projection

$$(y_1, \ldots, y_{n+1}) \mapsto (y_1, \ldots, y_n)$$

is itself a linear (though degenerate) transformation.

Note

$$\begin{pmatrix} y_1 \\ \vdots \\ y_n \\ 0 \end{pmatrix} = \operatorname{diag}(1, \ldots, 1, 0) \begin{pmatrix} y_1 \\ \vdots \\ y_n \\ y_{n+1} \end{pmatrix}. \tag{1.5}$$

More precisely, a Gaussian process is completely specified once we assign a mean value μ_i to every random variable y_i, and a covariance value Σ_{ij} to every pair (y_i, y_j). These quantities must satisfy the usual symmetry and positive semi-definiteness conditions; under these consistency requirements, they uniquely determine a Gaussian process.

We write a Gaussian process as

$$f \sim \mathcal{GP}(m(\cdot),\, k(\cdot,\cdot))\,, \tag{1.6}$$

where the mean function $m : \mathcal{X} \to \mathbb{R}$ is defined by

$$m(x) = \mathbb{E}[f(x)] := \int_{\mathcal{X}} f(x)\, \mathrm{d}\mu(x), \tag{1.7}$$

and the covariance kernel $k : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ is given by

$$k(x, x') = \mathrm{Cov}(f(x), f(x')) = \mathbb{E}[(f(x) - m(x))(f(x') - m(x'))] , \tag{1.8}$$

which must be positive semi-definite. In short, once the mean function and covariance kernel are fixed, the entire Gaussian process is already written; nothing further needs to be said.

Remark

In practice, we may assume that the kernel k is strictly positive definite. If the corresponding covariance matrix Σ is only positive semi-definite, we can add a small regularization term on the diagonal, for example

$$\Sigma_\varepsilon = \Sigma + \varepsilon I, \qquad \varepsilon > 0, \tag{1.9}$$

which ensures $\Sigma_\varepsilon \succ 0$. This is equivalent to adding a small independent Gaussian noise term to the process and is commonly used to avoid numerical degeneracy.

Example 1.1

In geostatistical terrain modeling, assume we observe discrete elevation samples

$$\{(x_i,\ y_i)\}_{i=1}^n, \qquad x_i \in \mathcal{X} \subset \mathbb{R}^2, \tag{1.10}$$

where each y_i denotes the measured height at location x_i.
To obtain a smooth terrain surface from these samples, we place a Gaussian process prior on the unknown height function $f : \mathcal{X} \to \mathbb{R}$:

$$f \sim \mathcal{GP}(0,\ k_{\mathrm{RBF}}(\cdot, \cdot)) . \tag{1.11}$$

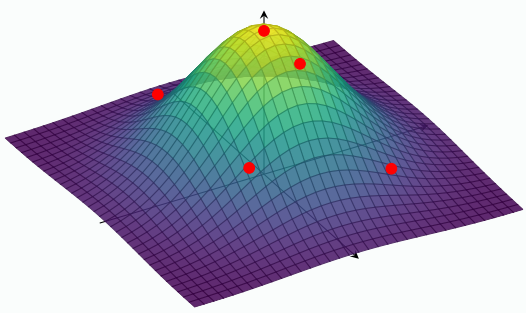
The radial basis function (RBF) kernel is given by

$$k_{\mathrm{RBF}}(x, x') = \sigma^2 \exp\left(-\frac{\|x - x'\|^2}{2\ell^2}\right), \tag{1.12}$$

which controls smoothness and spatial correlation of the terrain.
Conditioning on the observed elevations $\{y_i\}$, the posterior mean

$$m_f(x) = \mathbb{E}\left[f(x) \mid \{y_i\}_{i=1}^n\right] \tag{1.13}$$

yields a smooth interpolating surface, where the height at each location is a kernel-weighted combination of nearby observations.



Thus, the RBF kernel spreads the influence of each sample point smoothly across the region, producing a continuous terrain map consistent with the measured elevations.

In many applications, the mean function is set to be identically zero, $m(x) \equiv 0$, which corresponds to placing an uninformative prior on f. Under the assumption that the mean function is identically zero, $m(x) \equiv 0$, and given a kernel $k(x_i, x_j)$, the vector of training labels satisfies

$$y \sim \mathcal{N}(0, K), \tag{1.14}$$

where K is the Gram matrix defined previously, and the mean 0 denotes the zero vector in $\mathbb{R}^n$ with the same dimension as

y.

Let $\{x_i\}_{i=1}^n$ be the training inputs and $\{y_i\}$ their corresponding observations. The above Gaussian model describes the joint distribution induced by the Gaussian process prior on the training set. For a new test point $(x_\star, y_\star)$, we can augment the training vector to obtain the joint prior

$$\begin{pmatrix} y_\star \\ y \end{pmatrix} \sim \mathcal{N}\left(0, \begin{pmatrix} k(x_\star, x_\star) & k(x_\star, \cdot)^\top \\ k(x_\star, \cdot) & K \end{pmatrix}\right), \tag{1.15}$$

where

$$k(x_\star, \cdot) = \begin{pmatrix} k(x_\star, x_1) \\ \vdots \\ k(x_\star, x_n) \end{pmatrix}. \tag{1.16}$$

To make predictions, we require the conditional distribution $p(y_\star \mid y)$. Since the joint distribution is Gaussian and the marginal $p(y)$ is known, the conditional distribution can be obtained directly using the standard formula for conditioning a multivariate normal distribution.

Jumping directly to the conclusion, let

$$x = \begin{pmatrix} x_a \\ x_b \end{pmatrix}, \qquad x_a \in \mathbb{R}^a,\ x_b \in \mathbb{R}^b, \tag{1.17}$$

and assume

$$x \sim \mathcal{N}\left(\begin{pmatrix} \mu_a \\ \mu_b \end{pmatrix}, \begin{pmatrix} \Sigma_{aa} & \Sigma_{ab} \\ \Sigma_{ba} & \Sigma_{bb} \end{pmatrix}\right). \tag{1.18}$$

Then the following standard results hold:

- **Marginalization:**

$$x_a \sim \mathcal{N}(\mu_a, \Sigma_{aa}), \qquad x_b \sim \mathcal{N}(\mu_b, \Sigma_{bb}). \tag{1.19}$$

- **Conditioning:**

$$p(x_a \mid x_b) = \mathcal{N}\big(\mu_{a|b},\ \Sigma_{a|b}\big), \tag{1.20}$$

  where the conditional mean and covariance are

$$\begin{aligned} \mu_{a|b} &= \mu_a + \Sigma_{ab}\Sigma_{bb}^{-1}(x_b - \mu_b), \\ \Sigma_{a|b} &= \Sigma_{aa} - \Sigma_{ab}\Sigma_{bb}^{-1}\Sigma_{ba}. \end{aligned} \tag{1.21}$$

Remark

Intuitively, the matrix $\Sigma_{ab}\Sigma_{bb}^{-1}$ acts as a linear projection that describes how variations in x_b influence our estimate of x_a. It adjusts the mean of x_a in proportion to how far x_b lies from its own mean.

Returning to the main track, the predictive distribution for a test label $y_\star$ conditioned on the observed vector y is

$$p(y_\star \mid y) = \mathcal{N}(y_\star \mid \mu_\star,\ \Sigma_\star)\,, \tag{1.22}$$

where the posterior mean is given by

$$\mu_\star = k(x_\star, \cdot)^\top K^{-1} y, \tag{1.23}$$

and the posterior variance is

$$\Sigma_\star = k(x_\star, x_\star) - k(x_\star, \cdot)^\top K^{-1} k(x_\star, \cdot). \tag{1.24}$$

Assume that the underlying function values y are noise-free, and that in practice we only observe

$$\hat{y} = y + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, \sigma^2 I). \tag{1.25}$$

Under this observation model, one can show that the joint distribution of the latent value $y_\star$ at a test point $x_\star$ and the

observed vector $\hat{y}$ at the training inputs is given by

Theorem 1.2

$$\begin{pmatrix} y_\star \\ \hat{y} \end{pmatrix} \sim \mathcal{N}\left(0, \begin{pmatrix} k(x_\star, x_\star) & k(x_\star, \cdot)^\top \\ k(x_\star, \cdot) & K + \sigma^2 I \end{pmatrix}\right), \tag{1.26}$$

where K denotes the Gram matrix with entries $K_{ij} = k(x_i, x_j)$.

Proof

Notice that ε is independent of $(y_\star, y)$ and $\mathbb{E}[\varepsilon] = \mathbb{E}[y_\star] = \mathbb{E}[y] = 0$:

$$\begin{aligned}
\mathrm{Cov}(y_\star, y_\star) &= \mathrm{Var}(y_\star) = k(x_\star, x_\star), \\
\mathrm{Cov}(y_\star, \hat{y}) &= \mathrm{Cov}\big(y_\star,\ y + \varepsilon\big) \\
&= \mathrm{Cov}(y_\star, y) + \mathrm{Cov}(y_\star, \varepsilon) \\
&= k(x_\star, \cdot) + 0 = k(x_\star, \cdot), \\
\mathrm{Cov}(\hat{y}, \hat{y}) &= \mathrm{Cov}\big(y + \varepsilon,\ y + \varepsilon\big) \\
&= \mathrm{Cov}(y, y) + \mathrm{Cov}(y, \varepsilon) + \mathrm{Cov}(\varepsilon, y) + \mathrm{Cov}(\varepsilon, \varepsilon) \\
&= K + 0 + 0 + \sigma^2 I_n \\
&= K + \sigma^2 I_n.
\end{aligned}$$

■

Corollary 1.1

$$y_\star \mid \hat{y} \sim \mathcal{N}(\mu_\star, \Sigma_\star) \tag{1.27}$$

where $\mu_\star = k(x_\star)^\top (K + \sigma^2 I)^{-1} \hat{y}$.

Comparing the above predictive formulas with (**??**), we arrive at a striking conclusion:

Proposition 1.1

Linear regression, when viewed from the Bayesian perspective, induces a Gaussian process.

Proof

Take any finite collection of inputs $\{x_i\}_{i=1}^n \subset \mathcal{X}$ and define

$$f_i := f(x_i) = w^\top \phi(x_i), \qquad i = 1, \ldots, n. \tag{1.28}$$

Stack these values into

$$f := \begin{pmatrix} f_1 \\ \vdots \\ f_n \end{pmatrix} \in \mathbb{R}^n. \tag{1.29}$$

Let $\Phi \in \mathbb{R}^{n \times p}$ be the feature matrix

$$\Phi := \begin{pmatrix} \phi(x_1)^\top \\ \vdots \\ \phi(x_n)^\top \end{pmatrix}, \tag{1.30}$$

so that

$$f = \Phi w. \tag{1.31}$$

With the prior $w \sim \mathcal{N}(0, \sigma_w^2 I_p)$, f *is a linear transformation of a Gaussian vector, hence*

$$f \sim \mathcal{N}\big(0,\ \sigma_w^2 \Phi\Phi^\top\big). \tag{1.32}$$

Therefore, for all i, j,

$$\mathrm{Cov}\big(f(x_i), f(x_j)\big) = \big(\sigma_w^2 \Phi\Phi^\top\big)_{ij} = \sigma_w^2\, \phi(x_i)^\top \phi(x_j). \tag{1.33}$$

Define the kernel

$$k(x, x') := \sigma_w^2\, \phi(x)^\top \phi(x'). \tag{1.34}$$

Then for any finite set $\{x_i\}_{i=1}^n$, *the vector* $(f(x_1), \ldots, f(x_n))$ *is multivariate Gaussian with mean zero and covariance matrix* $[k(x_i, x_j)]_{i,j}$. *By the definition of a Gaussian process, this implies*

$$f \sim \mathcal{GP}\big(0,\ k(\cdot,\cdot)\big). \tag{1.35}$$

Now add Gaussian observation noise:

$$y_i = f(x_i) + \varepsilon_i, \qquad \varepsilon_i \sim \mathcal{N}(0, \sigma^2), \tag{1.36}$$

with $\{\varepsilon_i\}$ *i.i.d. and independent of* w. *Let* $y = f + \varepsilon$ *where* $\varepsilon \sim \mathcal{N}(0, \sigma^2 I_n)$. *Then* y *is also Gaussian with*

$$\mathrm{Cov}(y_i, y_j) = \mathrm{Cov}\big(f(x_i), f(x_j)\big) + \mathrm{Cov}(\varepsilon_i, \varepsilon_j) \tag{1.37}$$

$$= \sigma_w^2\, \phi(x_i)^\top \phi(x_j) + \sigma^2 \delta_{ij}. \tag{1.38}$$

Hence the effective covariance function for the observed process is

$$k_{\mathrm{obs}}(x, x') = \sigma_w^2\, \phi(x)^\top \phi(x') + \sigma^2 \delta_{x,x'}, \tag{1.39}$$

which is exactly the GP regression model with kernel as stated. ■

Remark

The GP framework is powerful because it allows us to work directly with the covariance kernel $k(x, x')$ *rather than introducing a weight vector* w. *This enables the kernel trick and makes infinite-dimensional feature spaces accessible without ever defining* $\phi(x)$ *explicitly.*

Remark

In practice, Gaussian process regression requires inverting the $n \times n$ *covariance matrix* $K + \sigma^2 I$, *which has a cubic time complexity* $O(n^3)$ *and becomes prohibitive when the dataset is large. To address this issue, one typically resorts to approximation techniques such as selecting a small set of inducing points or using low-rank and sparse approximations of the kernel matrix. These methods preserve most of the predictive power of the GP while significantly reducing computational cost.*

1.2 Gaussian Process Regression (GPR)

Review the previous conclusion. Given training inputs $X = \{x_i\}_{i=1}^n$ with observations

$$y = f(X) + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, \sigma^2 I), \tag{1.40}$$

and a test input $x_\star$, the joint distribution between y and the latent value $f_\star = f(x_\star)$ is

$$\begin{pmatrix} f_\star \\ y \end{pmatrix} \sim \mathcal{N}\left(0,\ \begin{pmatrix} k(x_\star, x_\star) & k(x_\star, X)^\top \\ k(x_\star, X) & K + \sigma^2 I \end{pmatrix}\right), \tag{1.41}$$

where

$$K_{ij} = k(x_i, x_j), \qquad k(x_\star, X) = \begin{pmatrix} k(x_\star, x_1) \\ \vdots \\ k(x_\star, x_n) \end{pmatrix}. \tag{1.42}$$

Conditioning on the observed vector y, the predictive distribution of $f_\star$ is Gaussian:

$$f_\star \mid y \sim \mathcal{N}(\mu_\star,\ \Sigma_\star), \tag{1.43}$$

with posterior mean

$$\mu_\star = k(x_\star, X)^\top (K + \sigma^2 I)^{-1} y, \tag{1.44}$$

and predictive variance

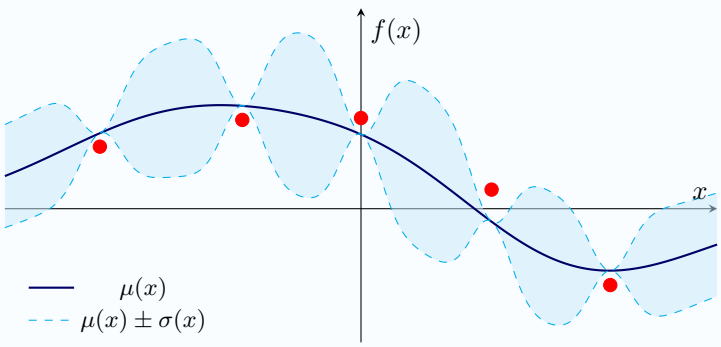
$$\Sigma_\star = k(x_\star, x_\star) - k(x_\star, X)^\top (K + \sigma^2 I)^{-1} k(x_\star, X). \tag{1.45}$$

Remark

Consider the RBF kernel

$$k_\ell(x, x') = \sigma_f^2 \exp\left(-\frac{\|x - x'\|^2}{2\ell^2}\right), \tag{1.46}$$

where $\ell > 0$ is the length-scale parameter.



For any fixed x, x', we have

$$\lim_{\ell \to 0} k_\ell(x, x') = \begin{cases} \sigma_f^2, & x = x', \\ 0, & x \neq x'. \end{cases} \tag{1.47}$$

In particular, for distinct training inputs $\{x_i\}_{i=1}^n$, the Gram matrix K_ℓ converges to a diagonal matrix:

$$\lim_{\ell \to 0} K_\ell = \sigma_f^2 I. \tag{1.48}$$

In the noise-free case (i.e. $\sigma^2 = 0$ in the observation model), the predictive mean at a training point x_j is

$$\mu_\ell(x_j) = k_\ell(x_j, X)^\top K_\ell^{-1} y. \tag{1.49}$$

In the limit $\ell \to 0$, we have

$$k_\ell(x_j, X) \longrightarrow \sigma_f^2 e_j, \qquad K_\ell^{-1} \longrightarrow \frac{1}{\sigma_f^2} I, \tag{1.50}$$

where e_j is the j-th standard basis vector in $\mathbb{R}^n$. Hence

$$\lim_{\ell \to 0} \mu_\ell(x_j) = \sigma_f^2 e_j^\top \left(\tfrac{1}{\sigma_f^2} I_n\right) y = e_j^\top y = y_j. \tag{1.51}$$

That is, in the limit $\ell \to 0$, the GP interpolates the data exactly at every training input: each training point is matched perfectly by the predictive mean.

From the discussion above, we know that the *posterior mean* of a Gaussian Process provides the predictive value for a test input, while the associated *uncertainty* can be quantified through the *predictive variance*. The relative standard

deviation is simply the square root of this predictive variance.

Gaussian Process Regression (GPR) serves as the mathematical foundation of **Bayesian Optimization (BO)**, a framework for performing optimization when the objective function is expensive, noisy, or lacks analytic structure.

Definition 1.6 (Black-box Optimization)

An optimization problem is called black-box optimization *if and only if the analytical form of the objective function is unknown and no gradient information is available.* ♣

Since the gradient of the objective function is unavailable, gradient-based methods such as gradient descent cannot be applied in black-box settings. The only feasible operation is *point-wise evaluation*: we may query the black box at a finite number of input locations and observe the corresponding outputs.

Bayesian Optimization (BO) aims to construct a probabilistic surrogate model of the black-box function and use it to locate the maximizer of y with as few evaluations as possible. This is particularly important when each evaluation is expensive, for instance in hyperparameter tuning.

The typical BO procedure proceeds as follows:

1. Randomly or uniformly select a small set of initial points $x_1, \ldots, x_n$ and obtain their evaluations $y_1, \ldots, y_n$.
2. Fit a Gaussian Process Regression (GPR) model using the collected data.
3. Use an acquisition function $a(x)$ to select one or a batch of new query points; evaluate them, augment the dataset, and refit the GPR model.
4. Repeat the process until the evaluation budget is exhausted, and return the point achieving the maximum observed value of y.

Here are two commonly used acquisition functions in Bayesian Optimization:

1. **Expected Improvement (EI)**:
$$a(x) = \mathbb{E}\big[(y(x) - y_{\max})^+\big], \tag{1.52}$$
where
$$[z]^+ := \max(0, z). \tag{1.53}$$
Under the GPR model, the predictive distribution is
$$y(x) \sim \mathcal{N}\big(\mu(x), \Sigma(x)\big), \tag{1.54}$$
so EI admits a closed-form expression obtained by integrating over the tail above $y_{\max}$:
$$a(x) = \int_{y_{\max}}^{+\infty} \big(y(x) - y_{\max}\big)\, \mathcal{N}(y(x) \,|\, \mu(x), \Sigma(x))\, \mathrm{d}y(x). \tag{1.55}$$
2. **Upper Confidence Bound (UCB)**:
$$a(x) = \mu(x) + \kappa\, \sigma(x), \tag{1.56}$$
where $\sigma(x)$ denotes the predictive standard deviation and $\kappa > 0$ controls the exploration–exploitation balance. In essence, UCB prefers points where the model is either promising (large μ) or highly uncertain (large σ).

Remark

Optuna *is a widely used automatic hyperparameter tuning library, recommended in class for practical Bayesian Optimization.*